MAST20030 Differential Equations Semester 2 Exam 2018

Differential Equations (University of Melbourne)



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THE UNIVERSITY OF MELBOURNE

Student Number

Semester 2 Assessment, 2018

School of Mathematics and Statistics

MAST20030 Differential Equations

Writing time: 3 hours

Reading time: 15 minutes

This is NOT an open book exam

This paper consists of 4 pages (including this page)

Authorised Materials

- Mobile phones, smart watches and internet or communication devices are forbidden.
- Calculators, tablet devices or computers must not be used.
- One A4 sheet, **handwritten only** on both sides, is permitted.
- No books, printed/photocopied materials or notes are permitted.

Instructions to Students

- You must NOT remove this question paper at the conclusion of the examination.
- You should attempt all questions.
- There are 6 questions with marks as shown. The total number of marks available is 100.

Instructions to Invigilators

- Students must NOT remove this question paper at the conclusion of the examination.

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Question 1 (10 marks)
In this question, we consider the inhomogeneous second-order linear ODE

$$t(t+1)\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (3t+1)\frac{\mathrm{d}y}{\mathrm{d}t} + y(t) = \frac{6\log t}{t}.$$

(a) Find a and b such that $(t+a)^b$ is a complementary solution. (*3 marks*)

(b) Use your complementary solution and reduction of order to find the general solution of this ODE. (*5 marks*)

(c) Determine the particular solution that also satisfies $y(1) = 1$ and $y'(1) = 0$. (*2 marks*)

Question 2 (20 marks)
This question concerns the following homogeneous second-order linear ODE:

$$2ty''(t) + y'(t) - y(t) = 0.$$

(a) Show that this equation admits a unique formal power series solution of the form

$$y_1(t) = \sum_{j=0}^{\infty} c_j t^j,$$

with $c_0 = 1$. Solve for the c_j and write the unique solution $y_1(t)$ in sum form. (*8 marks*)

(b) What is the radius of convergence of your formal solution? On which intervals of $\mathbb{R}$ does it converge *uniformly*? (*3 marks*)

(c) By quoting any relevant results from lectures, state whether your formal solution $y_1(t)$ is a genuine solution or not. [*You do not have to show that your solution is genuine.*] (*2 marks*)

(d) Compute the wronskian of $y_1(t)$ with another linearly independent solution $y_2(t)$. Use the result to show that $y_2(t)$ can be written in the form

$$y_2(t) = t^a \sum_{j=0}^{\infty} d_j t^j,$$

for some $a, d_j \in \mathbb{R}$. Normalise y_2 so that $d_0 = 1$ and substitute the first three terms of $y_1(t)$ into the wronskian equation in order to determine a, d_1 and d_2 explicitly. (*7 marks*)

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Question 3 (15 marks)

Consider the second-order linear IVP

$$y''(t) + 2ty'(t) + 2y(t) = 0, \qquad y(0) = 1, \quad y'(0) = 0. \tag{1}$$

In this question, you may bring limits and derivatives inside integrals, as long as it makes sense to do so.

(a) Show that $\lim_{s\to\infty}(\mathscr{L}f)(s) = 0$, as long as f is a function whose Laplace transform exists. (*2 marks*)

(b) Prove the following identity of Laplace transforms, assuming that the transform of f exists:

$$(\mathscr{L}tf(t))(s) = -\frac{\mathrm{d}}{\mathrm{d}s}(\mathscr{L}f)(s). \tag{*2 marks*}$$

(c) Apply $\mathscr{L}$ to both sides of the ODE in Equation (1) to obtain a first-order linear ODE for $(\mathscr{L}y)(s)$. Thereby, show that

$$(\mathscr{L}y)(s) = -\frac{1}{2}\mathrm{e}^{s^2/4}\int \mathrm{e}^{-s^2/4}\,\mathrm{d}s. \tag{*4 marks*}$$

(d) Write $\int \mathrm{e}^{-s^2/4}\,\mathrm{d}s = \int_0^s \mathrm{e}^{-\sigma^2/4}\,\mathrm{d}\sigma + C$ and evaluate the constant C using part (a). (*4 marks*)

(e) Show that $y(t) = \mathrm{e}^{-t^2}$ is the unique solution of Equation (1), hence use your previous results to write down an expression for $(\mathscr{L}\mathrm{e}^{-t^2})(s)$ in terms of $\int_0^s \mathrm{e}^{-\sigma^2/4}\,\mathrm{d}\sigma$. (*3 marks*)

Question 4 (15 marks)

Consider the 1-dimensional laplacian $\mathsf{L} = -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$, acting on a suitably nice vector space of complex-valued functions that satisfy the Dirichlet boundary conditions $y(0) = y(1) = 0$.

(a) Show, from first principles, that L is hermitian with respect to the inner product defined by

$$\langle f, g\rangle = \alpha\int_0^1 \overline{f(x)}g(x)\,\mathrm{d}x, \tag{2}$$

where α is any positive real number and $\overline{f(x)}$ indicates the complex conjugate of $f(x)$. What can we conclude about the eigenvalues and eigenfunctions of L? (*5 marks*)

(b) Determine the eigenvalues λ_n and eigenfunctions $y_n(x)$ of L explicitly, including the values taken by the parameter n. (*4 marks*)

(c) Compute α in Equation (2) so that all the eigenfunctions $y_n(x)$ have norm 1. Now prove that

$$f(x) = \sum_{n=1}^{\infty} b_n y_n(x) \qquad \Rightarrow \qquad b_n = \alpha\int_0^1 y_n(x)f(x)\,\mathrm{d}x. \tag{*3 marks*}$$

(d) Show that $f(x) = 1$ results in $b_n = \dfrac{4}{\pi n}$, if n is odd, and $b_n = 0$ otherwise. (*3 marks*)

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Question 5 (20 marks)

For reasons best left unsaid, Doctor Nefario is studying temperature changes in a thin insulated wire of length 1. Initially, the temperature is constant along the wire (with value 1 say), but Doctor Nefario has connected the ends of the wire to a diabolical machine that fixes the temperature of the endpoints to be $\mathsf{e}^{-\alpha t}$, where $t \geqslant 0$ is time. She has ultimate control over the parameter $\alpha > 0$.

(a) Write down an IBVP for the temperature $u(x,t)$ of the Doctor's wire. (*2 marks*)

(b) Write $u(x,t) = u_{\text{lin.}}(x,t) + u_{\text{rest}}(x,t)$, where $u_{\text{lin.}}(x,t)$ is chosen so that $\partial_x^2 u_{\text{lin.}}(x,t) = 0$ and it satisfies the inhomogeneous boundary conditions above. Solve for $u_{\text{lin.}}$ and show that $u_{\text{rest}}(x,t)$ satisfies the following IBVP with homogeneous boundary conditions:

$$\frac{\partial u_{\text{rest}}}{\partial t} = D\frac{\partial^2 u_{\text{rest}}}{\partial x^2} + \alpha \mathsf{e}^{-\alpha t}, \qquad u_{\text{rest}}(x,0) = u_{\text{rest}}(0,t) = u_{\text{rest}}(1,t) = 0. \tag{*4 marks*}$$

(c) Justify the following eigenfunction expansion for $u_{\text{rest}}(x,t)$:

$$u_{\text{rest}}(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin(\pi n x).$$

Determine IVPs for the $b_n(t)$ and solve them to obtain the following formal solution for $u_{\text{rest}}(x,t)$:

$$u(x,t) = -4\alpha \sum_{m=0}^{\infty} \frac{\sin[\pi(2m+1)x]}{\pi(2m+1)} \frac{\mathsf{e}^{-\alpha t} - \mathsf{e}^{-\pi^2(2m+1)^2 D t}}{\alpha - \pi^2(2m+1)^2 D}.$$

Consequently, write down a formal solution for $u(x,t)$. (*10 marks*)

(d) Doctor Nefario now tunes the parameter α to $\pi^2 D$. What happens to the formal solution for $u(x,t)$ in this limit? Find where setting $\alpha = \pi^2 D$ necessitates a change in the derivation of the solution. (*4 marks*)

Question 6 (20 marks)

The following "short answer" questions may all be answered independently of one another.

(a) Show that the general solution of a linear differential equation may be expressed as the sum of a general complementary solution and a single, arbitrarily chosen, particular solution. (*2 marks*)

(b) Calculate the matrix exponential $\mathsf{e}^{\mathsf{A}t}$, where $\mathsf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. (*4 marks*)

(c) The function given by $f(x) = \dfrac{1-x}{1+|x|}$, $-1 \leqslant x \leqslant 1$, has a Fourier series $g(x)$, $x \in \mathbb{R}$. Determine $g(n)$, for all $n \in \mathbb{Z}$. Would the finite sum approximations to this series exhibit Gibbs phenomena? (*3 marks*)

(d) Explain why a solution of Laplace's equation is never unique when all the (possibly inhomogeneous) boundary conditions are Neumann. (*2 marks*)

(e) Given the Fourier transform $(\mathscr{F}\mathsf{e}^{-|p|})(p) = \dfrac{2}{1+x^2}$, use convolution to compute $(\mathscr{F}\dfrac{1}{(1+x^2)^2})(p)$. (*5 marks*)

(f) Compute the Green function of the IVP $y'''(t) - y'(t) = f(t)$, $y(0) = y'(0) = y''(0) = 0$. (*4 marks*)

End of Exam—Total Available Marks = 100

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